

The Digital Asset Delivery Service (DADS)

from DXC Business Exchange Services (BES)

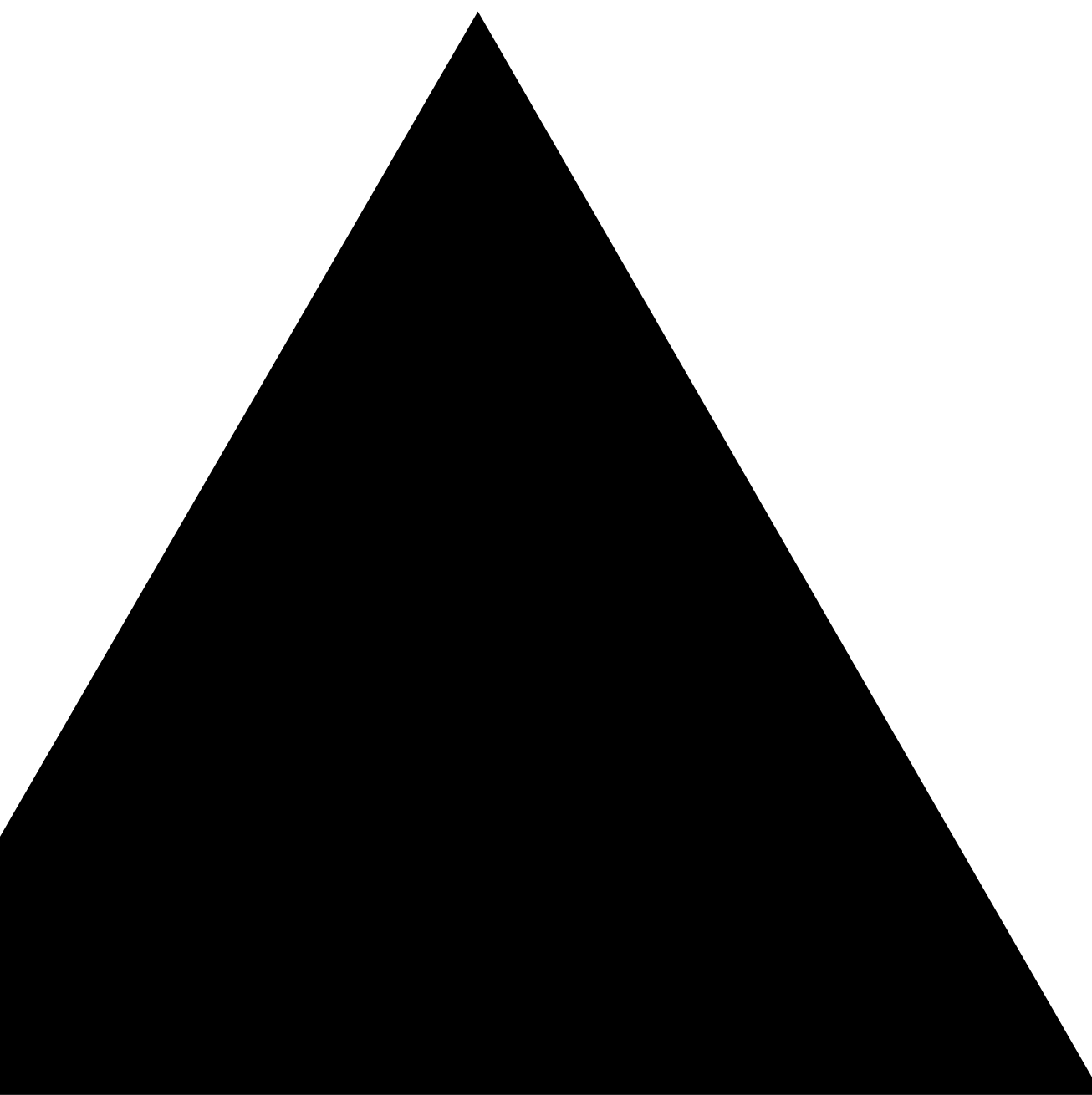


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Deployment

The system was rapidly deployed to over 750 secure managed endpoints and achieved 100% on-time on-quality deliveries. Our cloud based approach minimized cost and maximized scalability along with flexibility – permitting a ‘pay per use’ charging approach

DXC BES in Numbers

- 4 Processing Centers
- 1750+ Clients
- 135,000+ Trading Partners
- 3 Billion+ Transactions Annually

The DXC Digital Asset Delivery Service

This paper describes the Digital Asset Delivery Service (DADS) developed by DXC Business Exchange Services (BES) to provide a secure, efficient, traceable system to enable the management, monitoring and delivery of large digital assets against demanding deadlines over a continent wide heterogeneous limited bandwidth IP network.

Introduction

Drawing on its 35 years of experience in designing, deploying, securing and managing critical digital assets deliveries, DXC Business Exchange Services (BES) was asked to develop an end2end solution for the exchange of very large (up to 600GB) multi-part (over 1000 files) high quality very high value (~\$Billions) complex digital assets for the movie distribution industry. BES had to ensure that these assets could be delivered 100% on-time over limited bandwidth links to end points distributed over a continent with varying quality of links to them.

DXC BES developed a cloud based system to:

- Ingest, store and manage large digital assets
- Manage orders for the transfer of the assets
- Schedule the transfer of the assets
- Monitor, validate and report the transfer of the assets
- Monitor and secure the delivery end points for the assets

All within a secured scalable cloud environment.

Delivery Mechanisms

The BES DADS system can support a heterogeneous data transport environment, and has patented scheduling methodologies which optimize the delivery efficiency over those environments, exploiting current performance information.

The two key modes of delivery are

- Content Delivery Network (CDN) based
- Satellite based

A CDN based delivery solution is outlined in Figure 1.

Key Elements of BES DADS

- Asset Store
- Delivery Management
- Transfer Monitor
- Transfer Validate Security

BES DADS Asset Properties

- 600+ GB Size
- 1000s files per asset
- Values \$Billions
- Time Critical

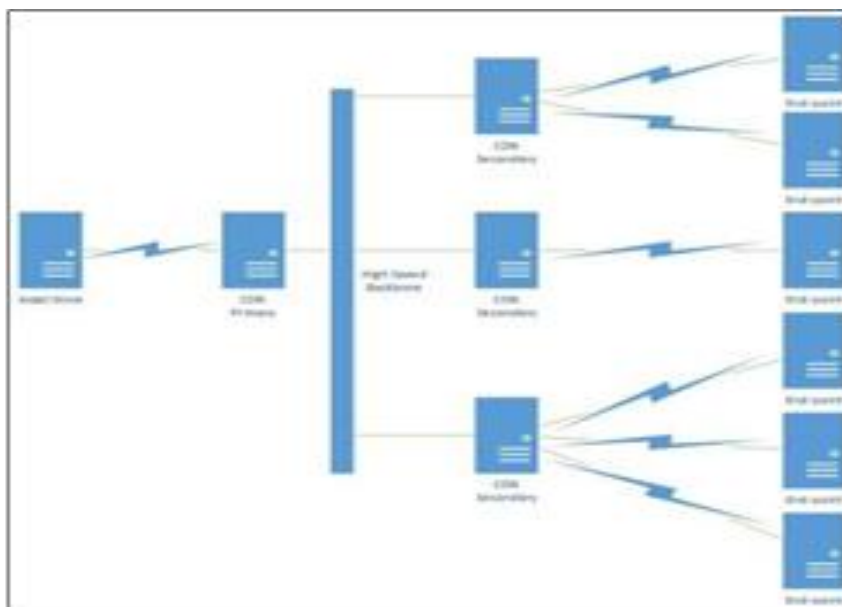


Figure 1. CDN based asset delivery system

The key in CDN based delivery is that asset transport is asynchronous and stored within the content delivery network. That is, each end-point can access different content at the same time with different transfer speeds, leading to complex scheduling and monitoring requirements. The structural difference in satellite based delivery is illustrated in Figure 2.

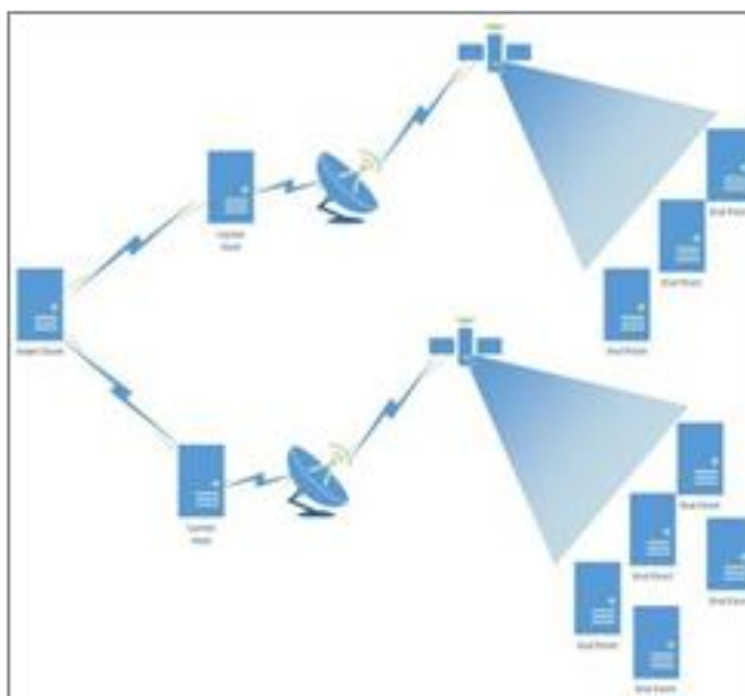


Figure 2. Satellite based content delivery system

The difference in satellite delivery is that the delivery is synchronous per satellite per channel and is sensitive, even with significant error correction overhead, to unpredictable weather interference – which may require either:

- Repeat transmission – inefficient
- Per site component retry – complex

However the broadcast nature of satellite does mean that transmission can be undertaken to many sites simultaneously.

DXC BES DADS is transport neutral and can optimize delivery using either or both network mechanisms.

Functional Architecture

DXC BES developed the key functional blocks to manage, monitor and report on the status of complex digital assets deliveries over a continent wide delivery network. The key functional blocks are illustrated in Figure 3.

Within the DXC BES DADS system DXC has direct responsibility for all of the functional elements apart from those under delivery.



Figure 3. Key functional blocks in BES DADS solution

Cloud Architecture

To minimize cost and maximize scalability along with flexibility DXC BES DADS has exploited a cloud based delivery system. An outline architecture for that system and its controlled access points is presented in Figure 4 which is simplified for clarity. Also included are development, staging, backup and disaster recovery environments.

Our cloud architecture allows us to implement 'pay per use' charging. The charging model for the DXC BES DADS solution is based on the volume of assets stored and the number of deliveries scheduled, managed and monitored.

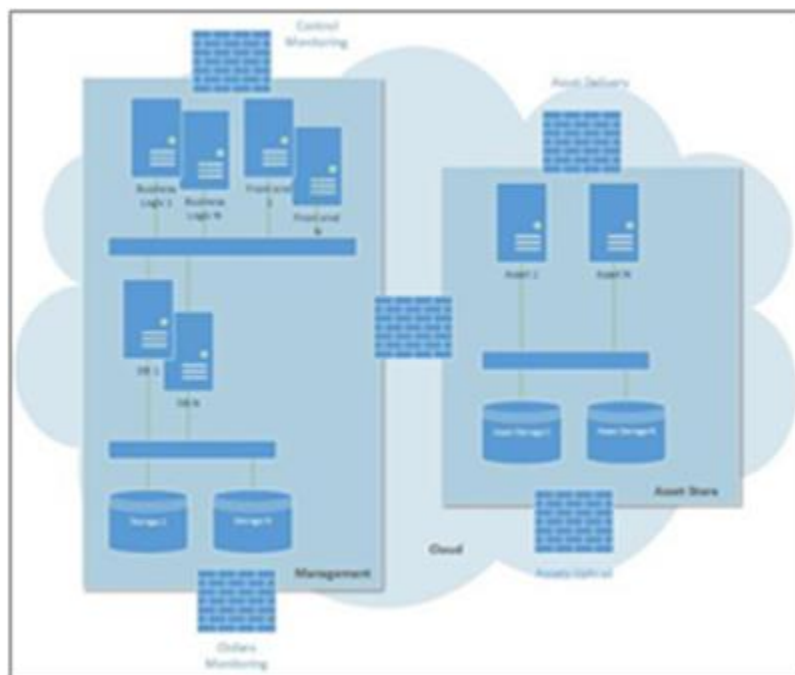


Figure 4. DXC BES DADS Cloud Architecture

Conclusion

On a 'pay per use' basis the DXC BES DADS system was rapidly deployed and scaled to over 750 end points in a continent wide delivery system. The DXC team successfully managed the 100% on-time delivery of all of the schedulable transfers within the system.



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Peter Lloyd-Brown managed the DXC BES Team that designed, created, deployed and managed all aspects of the DXC DADS system. Dr. Chris Tofts was the technical lead with over 30 years of performance, scaling and scheduling design experience.

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